Project proposal

● *The problem and the project scope*

Title: One Frame Is Enough: Miss-Aware Semantic Obfuscation with Adaptive Synthetic Identity Replacement on Edge Devices

1. introduction to project scope and the role of Edge AI within the context
   Our project focuses on protecting people's identities in videos while keeping the footage useful for analysis.We plan to detect and anonymize faces using blurring or synthetic identity replacement. The main challenge is handling frames where face detection fails, leaving someone's identity exposed.
   Edge AI plays an important role because video processing and anonymization happen directly on the device, without sending the original footage to the cloud.
   We also want to investigate how to balance privacy protection, processing speed, and limited hardware resources. This includes studying how model performance, detection frequency, and different anonymization techniques affect latency, FPS, and privacy risks on an Edge device.
2. challenges in traditional practices
   Most face anonymization systems depend on detecting faces correctly in every frame. However, faces can be missed because of poor lighting, movement, occlusion, or unusual angles.
   Even a single missed frame can expose someone's identity.
   Another challenge is that traditional blurring removes useful visual details, while realistic face replacement can be computationally expensive, especially on Edge devices.

3. assign roles and responsibilities within the team

Valerian — Computer Vision and Privacy

Face detection and tracking.

Recovery of missed face regions.

Face anonymization and synthetic feature replacement.

Privacy evaluation and testing.

Moro — Edge AI and System Integration

Model optimization and deployment on NXP.

Video processing pipeline.

Adaptive anonymization logic.

Performance testing (FPS, latency, memory).

4. project proposal ,problem statement and goals

Problem Statement

Current face anonymization systems can leave faces exposed when detection fails. Using stronger masking everywhere can solve some of these issues, but it also reduces video quality and usefulness.

Our Goal

We want to develop a system that uses information from previous frames and the surrounding scene to estimate where a missed face might be.Depending on how confident the system is, it will either replace the original facial features with a synthetic identity or apply safer masking when the location is uncertain.

Research Question

How can we use Edge AI to reduce privacy exposure caused by missed face detections while choosing the most suitable anonymization method and maintaining good video quality and real-time performance on limited hardware?

Expected Result

A working prototype on the NXP board, tested against standard face blurring and tracking-based anonymization, with comparisons of privacy protection, video quality, and processing speed.

● *Literature review*

1. **CIAGAN (CVPR 2020):** Uses generative models to replace a person's facial identity while keeping useful features, such as pose and expression. However, it does not focus on recovering missed face detections and choosing a safer fallback when replacement is unreliable.
2. **DeepPrivacy2 (WACV 2023):** Generates realistic anonymous faces and bodies. It mainly focuses on realistic anonymization rather than deciding between different privacy methods based on detection confidence and Edge hardware limitations.
3. **Face Anonymization Made Simple (WACV 2025):** Uses diffusion models to generate realistic anonymous faces while preserving expressions and head orientation. However, this approach is computationally demanding and does not specifically address missed detections on small Edge devices.
4. **SecurePose (2026):** Uses body pose information to estimate face locations when normal face detection is unreliable. This is useful for recovering missed faces, but it does not combine this recovery with realistic synthetic identity replacement and adaptive masking.
5. **Foresee and Protect You (2026):** Uses motion prediction and frame scheduling to reduce privacy exposure under processing constraints. However, it does not focus on realistic face replacement or selecting between different anonymization methods.
6. **LA3D (2024):** Adapts the level of video anonymization to balance privacy and video analysis performance. However, it does not specifically combine missed-face recovery with synthetic identity replacement and a conservative fallback.

Research Gap

From the literature we reviewed, we have not found a system that brings all these approaches together: recovering missed faces, replacing identities with realistic synthetic ones, and switching to safer masking when the system is uncertain or has limited processing resources.

Our idea is to combine these existing techniques into a hybrid anonymization system designed for Edge AI.

Instead of always applying the same method, the system would choose between synthetic identity replacement and conservative masking depending on how reliably it can locate the face and how much processing time is available.